

Question Paper

Exam Date & Time: 18-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Blockchain Architecture Design and Use Cases [NPT_4128]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Discover the lifecycle of chaincode deployment in Fabric from packaging to committing transactions. (3)
- A) Course Outcome: "CO5" / Cognitive Level: "Application"
- B) Identify how Fabric handles membership service providers (MSPs) and ordering nodes in a consortium network. (3)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- C) Analyze scalability and performance trade-offs between on-premise Fabric and IBM Blockchain Cloud implementations. (4)
- Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) Examine the sequence of chaincode lifecycle management - from installation to endorsement policy enforcement and instantiation on channels. (3)
- A) Course Outcome: "CO2" / Cognitive Level: "Analysis"
- B) Distinguish how private data collections enhance confidentiality in Fabric without compromising network transparency. (3)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- C) Design the deployment workflow of a Fabric network on AWS Managed Blockchain vis-à-vis manual setup using Docker and Kubernetes. (4)
- Course Outcome: "CO2" / Cognitive Level: "Create"
- 3) Evaluate how blockchain enables real-time cross-border payments and reduces settlement risk. (3)
- A) Course Outcome: "CO4" / Cognitive Level: "Analysis"
- B) Elaborate the use of smart contracts in automating trade settlements in global finance. (3)
- Course Outcome: "CO5" / Cognitive Level: "Create"
- C) Evaluate the impact of TradeLens on international logistics efficiency and transparency. (4)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"

4) Defend how blockchain reduces documentation delays in global trade logistics. (3)

A) Course Outcome: "CO4" / Cognitive Level: "Evaluate"

B) Classify the architectural flow of TradeLens and its key modules. (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

C) Criticize how blockchain networks like Hyperledger Fabric transform customs verification systems. (4)

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

5) Examine tokenization in trade finance and analyze how smart contracts automate Letter of Credit processes. (4)

A) Course Outcome: "CO3" / Cognitive Level: "Analysis"

B) Identify how TradeLens leverages IoT sensors and blockchain analytics to enhance maritime logistics visibility. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

C) Analyze the role of provenance data in ensuring ethical sourcing and authenticity in global supply chains. (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 NPT_4128:Blockchain Architecture Design and Use Cases

#	Course Outcome	Description	Marks
1	1	CO 1	7
2	2	CO 2	10
3	3	CO 3	21
4	4	CO 4	6
5	5	CO 5	6
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 NPT_4128:Blockchain Architecture Design and Use Cases

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	3	9(18%)
4	Analysis	8	27(54%)
5	Create	2	7(14%)
6	Evaluate	2	7(14%)
Total		15	50 (100%)